



Variability-based assessment of assisted gait using a multi-sensor instrumented cane

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ABSTRACT

The biomechanics of walking with assistive canes are not well understood clinically, despite their long history. Canes are also often misused or not used at all once taken home, despite the known benefits of proper use. To help clinicians and users evaluate and monitor cane use at home and in the community, a multi-sensor instrumented cane and interpretable gait performance metrics have been proposed. This study investigated the effects of an instrumented cane on gait speed, stride interval, and gait variability in 15 healthy individuals. Participants were asked to walk for 10 min under three different conditions: normal walking (as a control), walking with plantar flexion of the ankle, and walking with dorsiflexion of the ankle. The results showed that relying solely on speed-related gait features may not be sufficient to detect gait changes caused by both the use of a cane and the simulated gait conditions. To gain a more comprehensive understanding of gait performance, additional gait parameters based on gait variability extracted from the user's motion and/or the device's motion are necessary. Specifically, the detrended fluctuation analysis (DFA) exponent of stride-to-stride fluctuations can be used to differentiate between changes in walking and modifications in cane use, while other measures based on mean values, such as gait speed and stride interval, cannot. These results suggest that the abundance of information acquired from the inertial sensor-based instrumented cane can be used for automated gait analysis and gait recognition to observe and comprehend an individual's walking performance.

1. Introduction

Assistive ambulatory devices (AADs) are often used to help people with mobility limitations or impairments due to age, injury, or disease with gait training and rehabilitation. These devices are used to improve stability, reduce the load on the lower extremities, and generate movement, which can help prevent falls, injuries, or disability [1]. In Canada, more than a million people, representing 4.1% of the population, use walking aids, the most common of which are canes or walking sticks [2]. Canes have also been identified as the second most important AAD, after glasses, in a frail elderly population [3]. However, if used incorrectly, walking aids can lead to poor posture and musculoskeletal injury, as well as a 2.6-fold increase in the risk of falling in older adults [4]. Given the high prevalence of cane-assisted individuals and the potential for serious consequences, effective techniques are needed for the training and evaluation of cane use.

Physical therapists and rehabilitation specialists generally provide recommendations to cane patients based on qualitative indicators of gait performance, such as video recordings, observation-based scales,

and questionnaire-based scales. To improve the precision and objectivity of these assessments, quantitative techniques have been proposed, including force platforms for kinetic measurements and three-dimensional (3D) optical motion capture systems to measure spatio-temporal gait parameters and kinematics [5–8]. However, these costly processing systems are suitable only for laboratory research and cannot be used to evaluate gait patterns in real-world settings. Therefore, they are not practical for daily clinical practice for clinicians, caregivers, or users. As a result, there is a lack of quantitative information in the literature on the gait characteristics of people when using canes [5,6] despite a long history of cane usage.

Recent advances in wearable sensors and the Internet of Things (IoT) have allowed studies of human gait analysis that were traditionally performed with 3D motion capture systems and force plates to be conducted with more affordable sensors, such as inertial measurement units (IMUs) and strain gauges [9]. Such sensors are intended for environmental IoT applications and may be worn on the body of users (e.g., attached to their legs [10] or shoes [11]) or integrated directly into AADs [12–18]. Several studies have used wearable sensors or

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Fig. 1. An instrumented cane with IoT capabilities is equipped with strain gauges and a 9-axis IMU integrated into its handle.

instrumented AADs to monitor gait patterns during assisted walking, and can be divided into three categories based on the signals examined: (1) user movement (sensors attached to the user to measure lower limb movement) [10,11], (2) device movement (sensors attached to the device, that is, the cane) [12,18], and (3) both user and device movement [13–15]. Although IoT devices offer great improvements in terms of feasibility and application, this is a relatively new area of research in gait analysis, particularly with respect to (instrumented) cane use. As such, there are still unresolved issues in this field that need to be investigated.

To effectively track progress and inform interventions, clinicians and end users need *interpretable* gait performance metrics, which require the extraction of meaningful and descriptive features from raw sensor measurements. The features most commonly used for this purpose are the speed of walking and the stride time [7,19,20]. Principal component analysis of spatio-temporal parameters and joint kinematics of the lower extremities has revealed that gait speed is a major factor influencing overall spatio-temporal parameters [10]. However, there is no consensus on differences in gait speed with and without an assistive cane. For example, Martin et al. [10] found that the average gait speed for healthy subjects is slower when walking with assistance, while Bryant et al. [6] and Kloos et al. [21] observed the same in patients with Parkinson's disease (PD) and Huntington's disease (HD), respectively. On the other hand, Perez et al. [7] and Chong et al. [19] did not detect a significant difference in speed between unaided and cane-assisted walking for healthy subjects and patients with PD, respectively. Furthermore, although Perez et al. [7] found that five people with stroke increased their speed when using a cane compared to walking without assistance, a review of 12 experimental studies on cane use after stroke indicated conflicting differences in changes in gait speed (ranging from 0.09 m/s slower to 0.1 m/s faster using single-point canes, compared to unaided walking [22]). Therefore, speed-related gait parameters alone may not be suitable for monitoring gait changes after cane-assisted gait training and rehabilitation.

Fluctuations in movement during walking, known as *gait variability*, have been found to be more sensitive to gait changes and warning signs than measures based on mean values of walking speed [23]. This variability has been associated with older people [24], neurological disorders (such as PD, amyotrophic lateral sclerosis, and multiple sclerosis [25]), and falls [26]. To measure gait variability, two main characteristics are used [23,25,27]: (1) the magnitude of stride-to-stride fluctuations, usually either the standard deviation or the coefficient of variation (CV), and (2) the nonlinear dynamics of the stride-to-stride fluctuations, for example, the fractal scaling exponent using detrended fluctuation analysis (DFA) and the fractal dimension using Higuchi's method. However, few studies have evaluated cane-assisted gait using variability magnitude measures [7] and, to the best of our knowledge, no prior studies have evaluated the dynamic characteristics of cane-assisted gait variability.

The overarching goal of this study was therefore to compare state-of-the-art spatio-temporal gait performance measures (average gait speed and stride interval) and gait variability measures (CV and DFA of stride interval time series) for assessing and monitoring walking with the use of a cane. Towards this goal, the first objective of this study was to examine how the use of a cane impacts these four measures of gait performance as compared to walking without assistance, using body-worn sensors to track user movement. Using a custom multi-sensor instrumented cane, the second objective was to compare gait parameters obtained from the device's movement to those obtained about the user's movement from the body-worn sensors. The hypothesis was that the variability metrics obtained from these devices would provide further insights into user and device motion (i.e., better sensitivity and specificity to changes in gait and cane reliance) that are not apparent when using average spatio-temporal metrics, and would help clarify the discrepancies found in previous research that focused on parameters related to gait speed.

The third and final objective of this study was to evaluate the effectiveness of gait variability measures obtained from the instrumented cane for monitoring the effects of gait training and rehabilitation. To do this, the four gait parameters were evaluated as users walked normally and with simulated plantar flexion and dorsiflexion during cane use. It was hypothesized that the dynamic variability of *user movement* would reveal a high level of structure when walking normally (with and without the cane), while a drift towards randomness would be observed with altered gait conditions. In contrast, it was expected that *cane movement* would be more random during normal walking, as healthy participants would not have to rely on the cane for support. However, when walking with plantar flexion and dorsiflexion, we anticipated that the movement of the device would be less random due to an increased need for the participants' to support themselves. The findings of the three study objectives illustrate the advantages of the dynamic variability measure over typical performance metrics such as gait speed to assess and interpret *changes in gait* and *cane reliance*. Ultimately, this research presents a novel and efficient method of monitoring cane-assisted gait using an instrumented cane, with potential applications in gait training, rehabilitation, or long-term monitoring in clinic and community settings.

2. Methods

2.1. Multi-sensor instrumented cane

An IoT-enabled multi-sensor cane (Fig. 1) used in this study has been developed by the Health Technologies Laboratory (HTL) at the University of New Brunswick (UNB), Canada. Full details of the design and implementation of the system are presented in our previous research [28]. Briefly, the physical appearance of the instrumented cane is nearly indistinguishable from that of a traditional cane, but includes multiple sensors, data logging, and data streaming capabilities. It is equipped with a 9-axis IMU (LSM9DS1, STMicroelectronics) and strain gauges (SGT-1A/1000-TY13, Omega Engineering). The IMU produces readings from accelerometers, gyroscopes, and magnetometers, allowing for an accurate calculation of the cane orientation. In addition, the strain gauges, organized in a Wheatstone bridge configuration, can measure the load applied to the cane. Both the IMU and the strain gauges are sampled at 231 Hz. In this study, the instrumented cane was used to measure “device movement”.

To measure “user movement”, the Shadow[®] full-body motion capture system (Motion Workshop) was used. Shank-mounted inertial sensors were chosen from available sensors to acquire gait-related kinematic information with minimal motion artifacts [15]. These were positioned on the back side of the shank at a standard location midway between the heel and knee. The sampling rate of the Shadow system was set at 100 Hz.

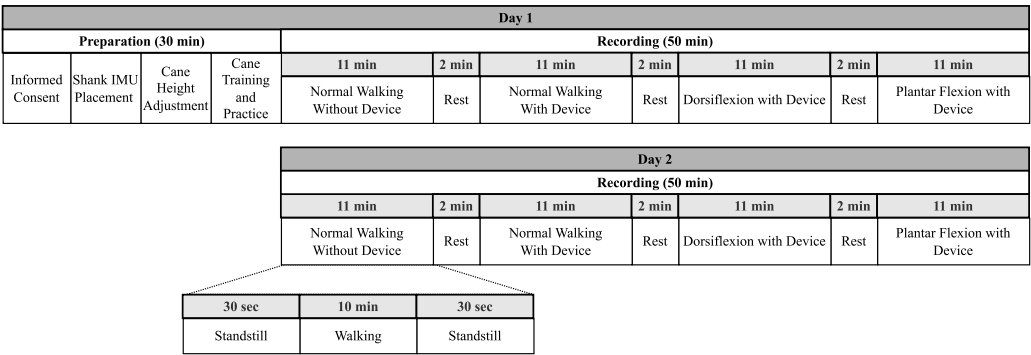


Fig. 2. Overview of the two-day data collection protocol.

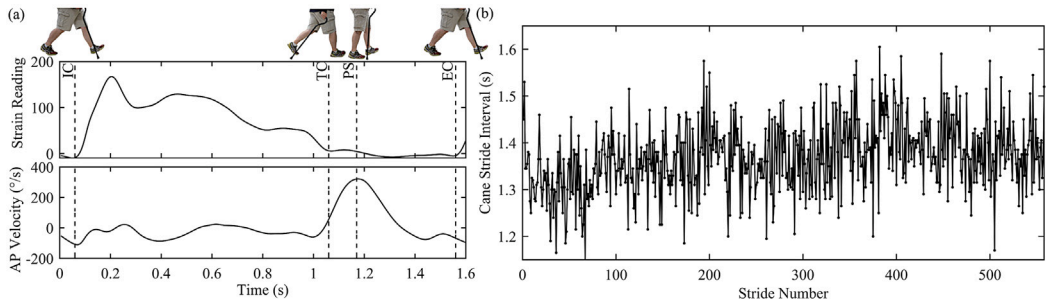


Fig. 3. (a) An example of cane measurements (strain reading and anteroposterior (AP) velocity) for one gait cycle, with initial-contact (IC), terminal-contact (TC), peak-swing (PS) and end-contact (EC, the IC for the next cycle) cane events indicated, and (b) an example of a stride interval time series from the cane for one walking trial (550 strides).

Table 1
Description of simulated gait conditions.

Gait condition	Description
Normal walking without device	Subjects walk without a cane on a level surface at their own chosen speed.
Normal walking with device	Subjects walk with a cane on a level surface at their own chosen speed. The cane was held on the opposite side of the left leg.
Dorsiflexion with device	Subjects walk with a cane on a level surface at their own chosen speed. The cane was held on the opposite side of the left leg, and they were instructed to keep their left toes off the ground by maintaining a dorsiflexed position.
Plantar flexion with device	Subjects walk with a cane on a level surface at their own chosen speed. The cane was held on the opposite side of the left leg, and they were instructed not to put any weight on the heel of their left leg by maintaining a plantarflexed position.

2.2. Data collection

Fifteen healthy individuals (12 men, 3 women, 18–31 years of age) with no known injuries participated in this study. All participants received a brief tutorial on cane-assisted walking (www.youtube.com/watch?v=fRn8ZZJMzno). The length of the cane was then adjusted to an appropriate height for each participant, ensuring that the handle aligned with their wrist when their arm was relaxed at their side [29]. This resulted in an elbow angle of approximately 15 to 20 degrees when the cane was held. The participants were given a practice session with the adjusted cane to become familiar with cane-assisted walking, and the experiment was only conducted once the participant felt confident enough to use the cane correctly. Each participant performed the walking trials listed in Table 1 (10 min for each test) with a standstill of 30 s while holding the cane off the ground at the beginning and end of each trial. On average, there were 550 strides per trial. The same trials were carried out on two consecutive days (day 1 and day 2), to strengthen measurement reliability and assess the short-term repeatability of the gait parameters. Fig. 2 summarizes the two-day data collection protocol. All subjects gave their informed consent to participate in the study, and the data collection procedures were supervised by the Research Ethics Board of the University of New Brunswick (REB 2017-097).

2.3. Data pre-processing

While the multi-sensor instrumented cane offers detailed data on loading, orientation, and movement, the current study concentrates on spatio-temporal gait characteristics obtained from detected stride intervals. Example cane measurements from one gait cycle are shown in Fig. 3(a). A gait cycle is identically defined for both the cane and the user, comprising (1) an initial-contact (IC) event that marks the start of the loading phase for the foot or cane, (2) a terminal-contact (TC) event that indicates the end of the loading phase, (3) a peak-swing (PS) event, occurring when the angular velocity of the cane/foot in the anteroposterior (AP) direction peaks as the user repositions the cane/foot for the subsequent loading phase, and (4) an end-contact (EC) event, serving as the IC event for the next cycle. A stride interval is defined as the time between the IC and EC events of a cycle. A multi-sensor matched filter algorithm was used to locate gait cycle events in the cane measurements [30]. After filtering the cane's strain gauge and IMU measurements using sixth-order low-pass filters with cutoff frequencies of 4 and 8 Hz, respectively, to eliminate high-frequency motion artifacts and other noise components (e.g., foot strike impact shock, vibrations, muscle tremors, small involuntary leg movements), the matched filter detected the relevant events in the AP gyroscope and strain readings based on their correlation to a template

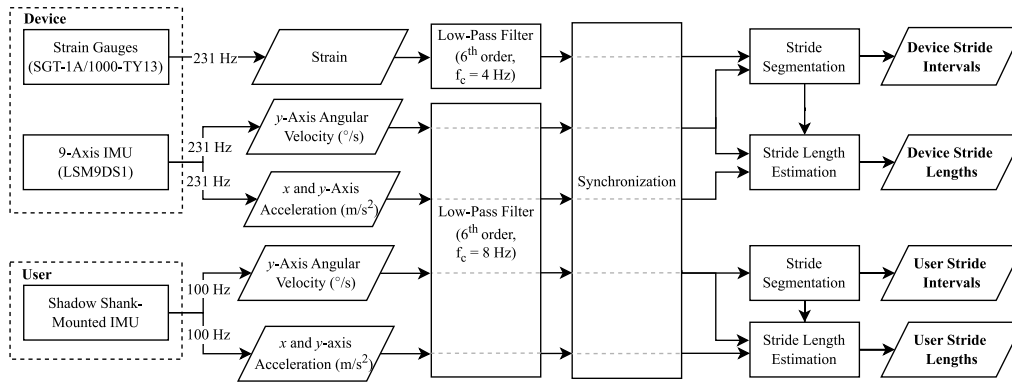


Fig. 4. Summary of the data pre-processing pipeline used to extract stride timing and length information from the raw sensor measurements.

gait cycle. This template was created by averaging ten manually-extracted strides from each participant, representing a typical stride for the individual. The matched filter segmentation procedure yielded the timing of IC events in the cane's sensor measurements, allowing the computation of stride interval time series for each trial (Fig. 3(b)). After segmentation, stride lengths were extracted from the cane recordings using an inverted pendulum model and knowledge of the length of the cane [31]. This approach is inspired by complementary filtering, and fuses information from the AP angular velocity and the acceleration in the AP and ML directions for better stride length estimation.

For the user's motion, stride intervals were derived from the IMU readings from the shank. Gyroscope measurements in the AP direction were pre-processed using a sixth-order low-pass filter with a cutoff frequency of 8 Hz, and a threshold-based peak detection algorithm was applied to identify gait events in the signal [15]. PS events were detected as the largest peaks in the AP angular velocity, and IC events were detected as the local minima between PS events. Similarly to the cane measurements, stride intervals were calculated as the duration between consecutive IC events. Stride lengths were determined through integration of the AP angular velocity and AP and ML accelerometer measurements as proposed by Sijobert et al. [32], using a drift correction technique to minimize accumulative integration error.

The beginning (and the end) of the two systems' recordings were synchronized by determining the first (and the last) positive strain gauge zero-crossing point in the cane data and the first (and the last) local minima in the AP angular velocity of the shank-mounted IMU. Fig. 4 provides an overview of the data pre-processing stages.

2.4. Spatio-temporal gait parameters

Gait speed and stride interval, also known as stride time and stride duration, are the predominant spatio-temporal parameters frequently used to assess the effectiveness of AADs [6,21] and continue to be a central focus of ongoing gait research [9,33]. Stride interval time series were obtained from both the segmented cane and the Shadow system recordings, based on the time between IC events. The average stride intervals of each trial were then used as key gait metrics for analysis. Likewise, the average gait speed was derived from the stride interval and the stride length time series for each trial.

2.5. Gait variability

Two metrics were used to quantify the variability of the stride intervals of the Shadow system (the user) and the instrumented cane (the device). The first is a static measure of the magnitude of stride-to-stride fluctuations, the coefficient of variation (CV). CV expresses the relative dispersion of the stride intervals, and is given by (1), where σ is the time series' standard deviation and μ is its mean.

$$CV = \frac{\sigma}{\mu} \times 100\% \quad (1)$$

The second measure, the detrended fluctuation analysis (DFA) exponent (α_{DFA}) [34], on the other hand, describes the fractal dynamics of the stride fluctuations by comparing the variability at different time scales. Briefly, a time series of length N is first integrated and then divided into L non-overlapping segments (or boxes) of equal length n . Order k polynomial least-squares fitting is then applied to each box, and the root mean square (or variance) of the detrended fluctuations is calculated according to (2),

$$F(n) = \sqrt{\frac{1}{N} \sum_{j=1}^N [y(j) - y_n(j)]^2}, \quad (2)$$

where y is the cumulative sum of time series and y_n represents the series of semi-local trends given by the fitted polynomials L . These steps are repeated for multiple box sizes n , and the DFA scaling exponent α_{DFA} is calculated as the slope of a straight line fit to the log-log graph of $F(n)$ and n . A value of $\alpha_{DFA} = 0.5$ implies the absence of long-term correlations (i.e., randomness). If $0 < \alpha_{DFA} < 0.5$, then it suggests that there are anti-persistent long-term correlations (i.e., large values are likely to be followed by small values and vice versa). On the other hand, if $0.5 < \alpha_{DFA} < 1$, it implies that there are persistent long-term correlations (i.e., large values are likely to be followed by large values, and small values are likely to be followed by small values). In this study, a detrending polynomial of degree two was used, with the minimum and maximum box sizes set to four and half of the total number of strides, respectively. The progression of box sizes followed a geometric pattern with a common ratio of two [27]. The first 512 steps of each walking trial were used for the CV and α_{DFA} calculations. Examples of the four gait measures, plotted over time for visualization purposes, are shown in Fig. 5.

2.6. Statistical analysis

Before comparing two groups, a Lilliefors test was performed to determine whether the data from each group were normally distributed. If the null hypothesis that the data was from a normally distributed population was rejected ($p < 0.05$), the Wilcoxon signed rank test was used to compare the two conditions. If the null hypothesis was not rejected, a paired t -test was used. Cohen's d was used to measure the effect size, and only tests that had both $p < 0.05$ and $d > 0.2$ were considered significant. Generally, an effect size of 0.2 is considered small, 0.5 is considered medium, and 0.8 is considered large.

3. Results

Two key comparisons were made to address the three objectives of the study. First, the Shadow system was used to compare the spatio-temporal parameters (gait speed, average stride interval) and variability parameters (CV of stride intervals, α_{DFA} of stride intervals)

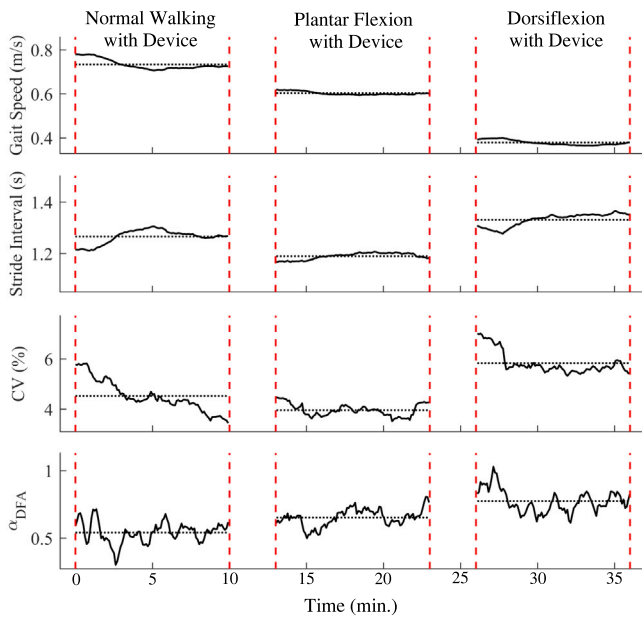


Fig. 5. Illustration of the four gait performance measures derived from cane measurements from one participant's session, computed using a sliding window of 128 strides. The participants performed the walking trials in sequence, with rests in between. Note: although the features are depicted here as time series over each 10 min walking trial, summary values were used for all statistical comparisons (one value per trial).

between two control trials, one with and one without the cane. This was to assess the effect of cane use on walking compared to unassisted walking, and the results are presented in Fig. 6. Significant decreases ($p < 0.05$) in gait speed were observed in users walking with a cane compared to without one ($d = 2.29$ on day 1; $d = 1.03$ on day 2). Furthermore, the average stride intervals increased significantly when using the cane ($d = 2.48$ on day 1; $d = 1.86$ on day 2). The static measure of variability, CV, of the stride intervals was also found to be significantly higher ($d = 0.87$ on day 1; $d = 0.78$ on day 2). The magnitude of the differences (d) between the walking trials with and without the cane was smaller on the second day, probably due to the fact that the participants became more accustomed to the cane. Although there were also differences between days in unaided normal walking for both gait speed ($d = 2.88$) and stride interval ($d = 0.37$), the trends remained consistent. By contrast, no differences were found in the dynamic measure of variability, α_{DFA} , of stride intervals. The median DFA exponents for unassisted walking were 0.90 and 0.87, and for assisted walking they were 0.94 and 0.90 on day 1 and day 2, respectively. These high α_{DFA} values indicate the presence of persistent long-term correlations in stride interval fluctuations.

Second, to explore the differences between user and device movement during cane walking, the four gait parameters were compared between the Shadow system (user) and the cane (device) for three different walking conditions: normal walking (as a control), walking with plantar flexion of the ankle, and walking with dorsiflexion of the ankle, and the results are presented in Fig. 7. Although the gait speed and stride interval parameters, measured from both the user and the device, remained the same in all situations, the variability parameters varied significantly when walking regularly and with plantar flexion. The stride interval CVs of the device were significantly higher than those of the user in both normal walking ($d = 1.80$) and plantar flexion ($d = 1.24$). Furthermore, the α_{DFA} exponents were significantly lower for the device than for the user in normal walking ($d = 4.37$) and plantar flexion ($d = 2.87$), indicating that the cane strides were more random than the strides of the lower extremity of the user. In the most difficult walking condition known as dorsiflexion walking, no differences in variability were observed between the stride intervals of the user and the

device. This implies that users relied more heavily on the cane during these trials and better integrated the device into their walking rhythm.

Fig. 7 also demonstrates the ability of spatio-temporal and variability parameters to monitor alterations in walking and the user's reliance on AADs. The gait speed was found to be significantly faster when walking normally than when performing plantar flexion ($p < 0.05$ and $d = 0.63$ for both the user and the device) and dorsiflexion ($p < 0.05$ and $d = 1.32$ for both), and the gait speed was higher during plantar flexion than during dorsiflexion ($p < 0.05$ and $d = 0.79$ for both). Likewise, the mean stride intervals were found to be significantly higher during dorsiflexion tests than during normal walking ($d = 0.70$ for the user; $d = 0.68$ for the device) and plantar flexion ($d = 0.85$ for the user; $d = 0.73$ for the device), and the CVs of the stride intervals were also higher during dorsiflexion than during normal walking ($d = 1.67$ for the user; $d = 0.49$ for the device) and plantar flexion ($d = 1.42$ for the user; $d = 0.44$ for the device). Interestingly, in contrast to the other gait parameters, opposite trends were observed between user movement and device movement for the α_{DFA} of stride intervals. As hypothesized, the DFA exponents revealed a shift towards randomness for the user, but a shift towards structure (long-term correlations) for the device as the participants' walking conditions were increasingly altered, significantly so for the dorsiflexion trials compared to normal ($d = 1.32$; for the user, $d = 2.37$ for the device) and plantar flexion trials ($d = 1.08$ for the user; $d = 1.64$ for the device). This finding not only confirms the increased reliance on the cane during dorsiflexion-altered walking trials, but also highlights the limitations of the traditional static gait parameters (such as gait speed) in evaluating cane use. This is due to the fact that these cane parameters follow the same trends as the user parameters, making it impossible to differentiate between changes in gait and changes in cane use.

4. Discussion

4.1. Impact of cane on gait

It has been widely reported that impaired gait is characterized by slower walking speed, longer stride time intervals, and increased gait variability (i.e., CV) [24]. AADs such as canes are often prescribed to provide additional stability and improve gait performance, with the expectation of increased walking speed, reduced stride time intervals, and decreased gait variability. Unfortunately, studies investigating the effect of walking aids on gait in new users have shown that gait performance remains unchanged or even worsens [6,21]. For example, Kloos et al. [21] compared the spatio-temporal gait parameters of HD individuals walking in a straight line and around obstacles with and without the use of AADs such as canes and walkers. The results revealed that the subjects walked significantly slower and with more variability when using AADs compared to walking unassisted. Similarly, Bryant et al. [6] studied gait changes in individuals with PD and found that subjects walked significantly slower when using AADs than without assistance, while stride intervals remained unchanged. Interestingly, similar results were found regardless of whether (1) no professional guidance was given to the subjects on how to use the device, which resulted in immediate gait changes when using a cane [6] or (2) the subjects were trained by a therapist and allowed to practice until they had a good understanding of the correct technique and felt comfortable using the device before data collection [21]. Consistent with two aforementioned studies conducted on patients [6,21] and previous research on healthy individuals [10,35], this current study revealed that inexperienced healthy individuals walked more slowly with a larger mean stride interval and a higher variability of stride intervals (CV) when using a cane compared to not using one (Fig. 6).

Previous research involving people who are used to walking aids has shown an improvement in gait performance in a variety of gait parameters [7,36–38]. For example, Polese et al. [36] found that people with chronic stroke who had used walking sticks for an average

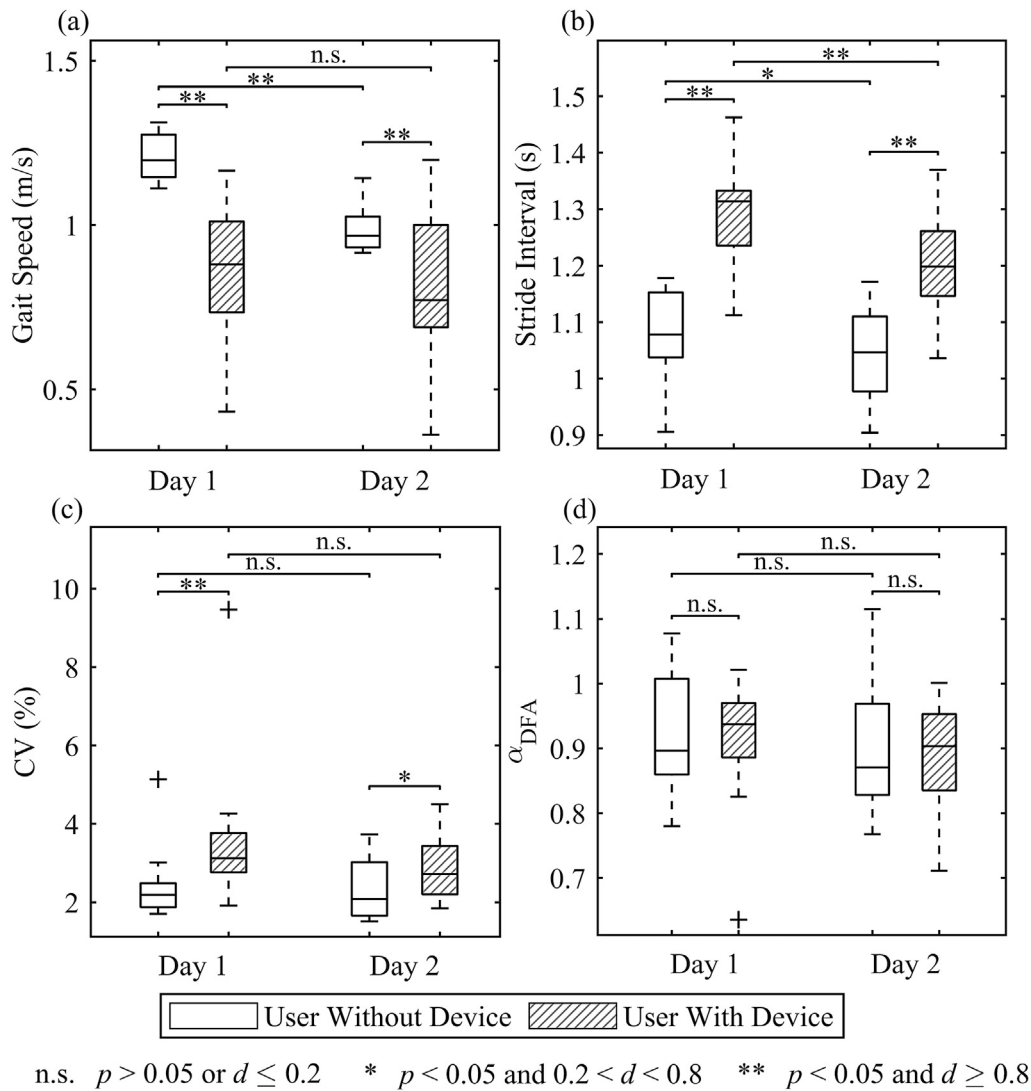


Fig. 6. Comparison of unassisted and assisted walking and between-day differences for four gait parameters derived from the user (Shadow system) sensor recordings during normal walking: (a) gait speed, (b) stride interval, (c) CV, (d) α_{DFA} .

of 6 years experienced faster walking speeds when using a cane. In a prospective longitudinal study conducted by Schwenk et al. [37], the effectiveness of a rollator walking aid was evaluated in geriatric patients (including those with hip fracture, respiratory disease, cardiac disease, and stroke) at the beginning and end of a 3-week rehabilitation program. The results showed that when using the aid, the walking speed increased and the stride time decreased compared to not using the aid, both pre- and post-testing. Hardi et al. [38] conducted a study on gait of elderly people in the community who had been using a walking aid for at least a month prior to their gait analysis. They discovered that stride-to-stride variability decreased when walking with a cane. However, they did not observe a significant improvement in gait speed when using a cane [38].

Self-selected walking speed is generally accepted as a good measure of overall gait performance [39,40]. However, healthy adults can walk at different speeds from day to day. In the current study, there were no differences between days for the gait variability measures (CV and DFA of the stride intervals), but the gait speed (without a cane) and the average stride intervals (with and without a cane) showed significant differences even when tested one day apart (Fig. 6). Furthermore, it has been observed in dual-tasking studies that when healthy young and old adults are asked to perform a secondary task while walking, their gait speed decreases [41]. However, the variability of stride and

swing intervals remains the same, even when the individuals are asked to walk and perform a challenging cognitive task (e.g. subtracting 7's in sequence). It appears that using only speed-related gait parameters may not be sufficient to monitor gait alterations resulting from gait training and rehabilitation. Therefore, additional gait parameters, such as gait variability, must be taken into account to obtain a comprehensive understanding of gait performance.

4.2. User movement vs. Device movement

Human gait is determined not only by the lower limbs, but also involves the coordination of the upper and lower limbs. Previous studies [42,43] have revealed a connection between arms and legs when walking normally, known as inter-joint coordination or inter-limb synergies. When healthy adults walk without an AAD, their arms move out of phase with each other at a rate that is in sync with the step frequency (i.e., two swings of the arm to one swing of the leg) at low walking speeds (0.3–0.7 m/s) and is in harmony with the stride frequency (i.e., one swing of the arm to one swing of the leg) at higher walking speeds (>0.8 m/s) [44,45]. This suggests that there are different coordination patterns within the human walking mode.

The use of a cane appears to activate the arm more in walking coordination, and the cane is effectively incorporated into synergies

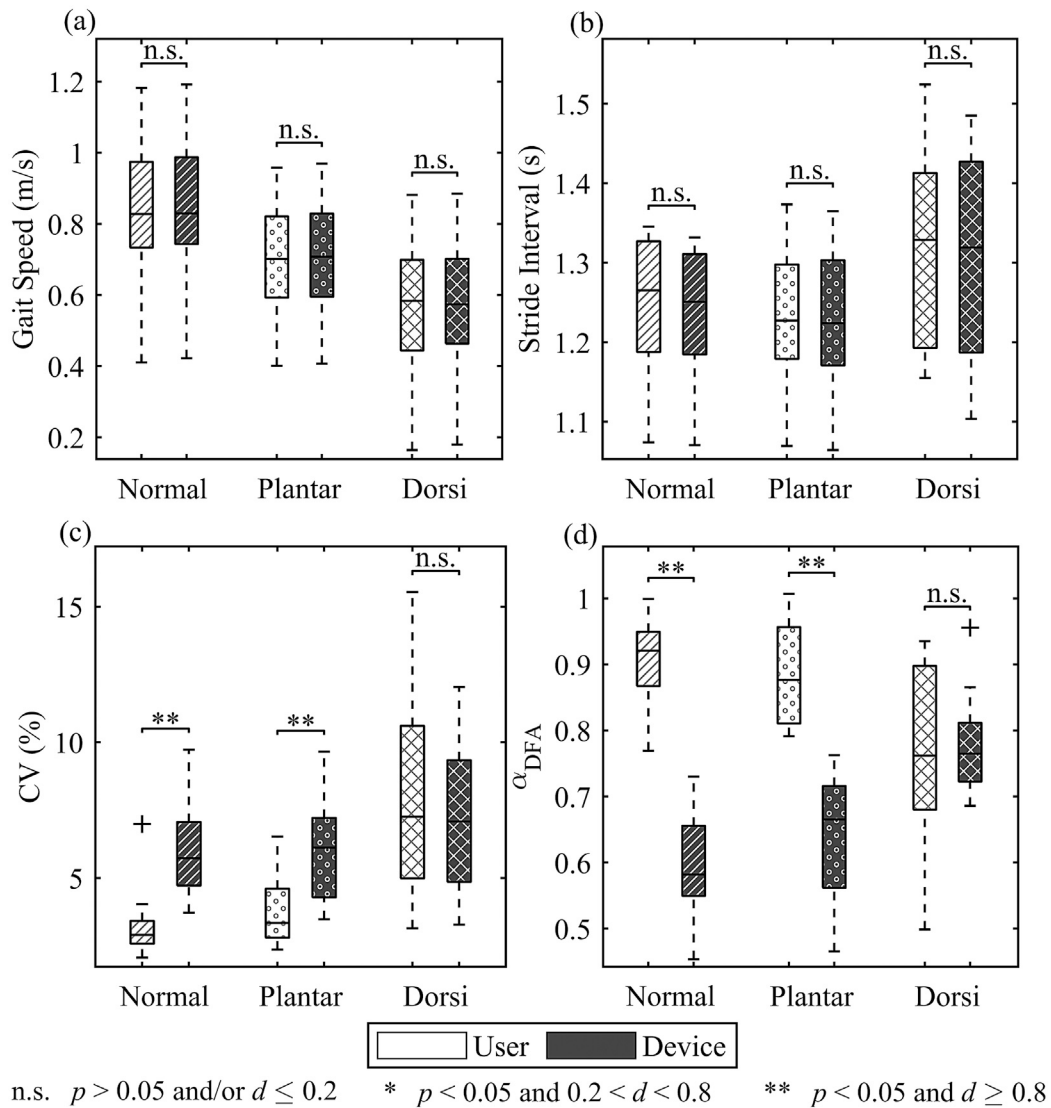


Fig. 7. Comparison between gait parameters derived from the user (Shadow system) and device (smart cane) sensor recordings for normal, plantar flexion, and dorsiflexion walking with the device: (a) gait speed, (b) stride interval, (c) CV, (d) α_{DFA} .

between the limbs [13], which could modify arm movement patterns. In practice, most users hold the cane on the opposite side and move it forward in sync with the affected lower limb, followed by the unaffected lower limb. This is known as the two-point gait. Sprint et al. [15] used stance and swing phase plots to confirm that the cane was in sync with the affected leg of stroke patients. Ribeiro et al. [46] also found a strong correlation between the inertial data of the foot and the cane for the mediolateral (ML) angular velocity and the vertical acceleration. They suggested that the cane pattern mirrors human gait indirectly, and thus gait events (e.g., heel strike, toe off, midstance, and midswing) can be identified using features extracted from cane movement with a convolutional neural network and long short-term memory (CNN-LSTM) (accuracy > 96%) [46]. In this study, this link allows us to gauge comparable mean values of spatio-temporal parameters from both the user's motion and the device's motion (Fig. 7(a–b)).

It is noteworthy that a large proportion of older cane users (≥ 65 years) do not use them correctly [47]; approximately 28% of these users hold the cane on the wrong side, 11% swing it with the leg on the same side as the cane, and 14% lift it up for multiple steps. Fortunately, instrumented canes can be beneficial for detecting incorrect cane usage. For example, if the cane is lifted off the ground for multiple steps, which means that the user is not using it for support, there would be no ground contact event within the gait cycle [15,28].

In comparison to the average spatio-temporal parameters that may be obtained from either the user or the cane, the gait variability of spatio-temporal parameters derived from the user's motion and the device's motion may be different (Fig. 7(c–d)). Consistent with the present results, Ribeiro et al. [46] discovered a minimal correlation in AP velocity, which was used to compute spatio-temporal parameters in this investigation, between the cane and the foot. The following subsection discusses the potential for these differences to be used for gait monitoring.

4.3. Gait monitoring

This research showed that α_{DFA} of the stride-to-stride fluctuations can be used to differentiate between alterations in walking patterns and adjustments in the use of a cane (reliance on assistive devices) (Fig. 7(d)), while none of the other measures based on mean values, such as gait speed and average stride interval, can. Gait variability can be seen as a reflection of the inconsistency in the central neuromuscular control system's ability to regulate gait and maintain a consistent walking pattern [23]. Thus, it is reasonable to assume that changes in gait variability would be associated with instability, as demonstrated by α_{DFA} moving closer to randomness as participants' walking conditions were increasingly altered (Fig. 7(d)).

Although gait variability observed by analyzing a person's movement can be used to determine whether they are walking in a consistent manner, it cannot be used to directly measure the efficacy of cane use. In the current study, when healthy young adults walked with a cane without any simulated conditions, their cane movements were random ($\alpha_{DFA} \approx 0.6$), as they were not heavily dependent on the cane. However, their lower extremity movements still showed long-range correlations ($\alpha_{DFA} \approx 0.9$). When the participants walked with their ankles flexed, they had to rely on their cane to keep their gait consistent and prevent them from falling. This caused the variability of their gait, measured by the dynamic movement of the cane, to become more structured (less random) ($\alpha_{DFA} \approx 0.7$ – 0.8). As a result, the α_{DFA} metric can be used to determine whether the cane is being misused or not used at all. Therefore, measuring dynamic stride-to-stride fluctuations of both the user and the device with α_{DFA} would allow us to observe any gait changes that may occur due to modifications in cane use.

However, it may be unwanted or infeasible to rely on body-worn sensors in unstructured, real-world settings, particularly for persons with physical or cognitive impairments. If one is looking to monitor daily activities without body-worn sensors, we can substitute the gait speed obtained from cane motion for the α_{DFA} derived from user motion, in combination with the α_{DFA} obtained from cane motion, to track alterations in gait when using a cane. Furthermore, while the current research concentrated on analyzing gait features based on stride intervals, the instrumented cane has the capacity to collect extra data concerning the utilization of the cane, including its position, speed, and the force it supports. Such data may be used to complement the spatio-temporal parameters and variability measures, to gain additional insight into cane dependence and to assess the strain on affected joints [48], detect anomalies such as tremors [49], or predict fall risk [18]. Indeed, strain and velocity measurements from the custom cane used in the present study have also demonstrated the ability to detect changes in gait and terrain during cane use [17].

The findings of the present study illustrate the utility of fractal-based variability analysis, a well-established method for measuring gait instability, for detecting changes in cane-assisted gait with an instrumented cane. Being the first study to investigate the non-linear dynamics of cane-assisted walking, this work presented an interpretable and effective framework for monitoring gait and cane reliance through the randomness or persistence of stride intervals. As the reliance on assistive devices is important to consider when evaluating the potential to achieve functional community ambulation [50], fractal-based variability analysis could serve as a tool for healthcare professionals, along with conventional measures such as gait speed and cane offloading, to help design training programs or other interventions involving the use of a cane. Due to the widespread incorrect use of canes and the limited knowledge in the research community about how assistive devices affect walking, the suggested approach could enhance the assessment of walking with a cane for rehabilitation and health monitoring in clinical or community settings.

4.4. Limitations and future work

The current study investigated immediate gait modifications in healthy individuals when using a single-point cane, but long-term effects for regular cane users in everyday and outdoor settings remain unknown. It is recognized that there are certain limitations with the present work, and more research needs to be done.

First, the subjects in the current study had not used canes regularly, so walking with canes was a new experience for them. It is likely that some changes in gait performance were due to the participants' unfamiliarity with the assistive device. Previous studies have also not reached consensus on differences in gait parameters over a period of cane use or training [36–38], and canes are frequently considered to be a confounding factor in gait analysis [51]. As training and periodic reassessment are essential for ensuring safe and effective use of walking

aids, future research should consider cane use over a longer period of time (weeks, months, or even years) to observe both immediate changes in gait and long-term effects for regular cane users.

Second, this study focused on a cohort of young healthy individuals and examined alterations in their walking patterns when two different simulated gait abnormalities were introduced. The results of this study were similar to those of previous studies conducted in older adults and patient populations [6,10,21], however, more research needs to be carried out to explore the differences related to age and gait disorders in gait biomechanics. Future works should re-evaluate the proposed variability-based analysis in target populations, especially over a long-term rehabilitation or monitoring period, and compare the obtained measures to gold standard clinical measures (e.g., from observation-based scoring by trained clinicians).

Third, the canes used in this study were tested in a laboratory setting, similar to many previous studies (e.g. [21,36,38]). This restricts the capacity to completely evaluate the device's performance in all possible scenarios. Since the instrumented cane used in this study can store data on a microSD card and use Bluetooth Low Energy to transmit data to a nearby smartphone or tablet, future research should investigate the use of canes in real-world and outdoor settings.

Fourth, in this study, a single-tip support cane was used, as it is a commonly used AAD. Other walking aids, such as quad canes, rolling canes, two-wheeled walkers, and four-wheeled walkers, are also frequently used by elderly individuals and those with gait impairments. It is possible that the gait changes produced by these devices may be different from the results of this study [11,21]. A new system design for various instrumented assistive devices may be necessary.

Finally, the DFA exponent of stride-to-stride fluctuations during walking can be used to measure gait variability. The results of this study are consistent with the idea that variability regulation is usually automated [23] and does not require much cognitive effort (i.e., for the healthy participants in this study, no differences were observed when walking with and without a cane). However, when automaticity is impaired (e.g., in the presence of age-related pathologies such as Alzheimer's disease [52]), cognitive tasks can affect gait variability. This study only looked at the effect of using a cane (with vs. without a cane) for normal walking, so future research should investigate the impact of a cane on gait variability during various perturbed walking conditions.

5. Conclusions

In conclusion, this research has highlighted the impact of cane use on walking speed, stride interval, and static and dynamic aspects of gait variability, which were obtained from the user's movement and the device's movement. It has been established that going beyond the first moment (the mean) and the second moment (the standard deviation) can be beneficial in understanding gait alterations due to cane use, by measuring dynamic stride-to-stride fluctuations from both the user and the device with the α_{DFA} exponent. The gait parameters extracted from the instrumented cane alone (such as the gait speed and α_{DFA} exponent measured from the cane's movement) also allow us to observe any gait changes that may occur due to modifications in cane use. This research demonstrated that a cane with embedded inertial and strain sensors can be used instead of body-worn sensors to detect changes in walking and assess the reliance and misuse of the cane.

CRediT authorship contribution statement

Angkoon Phinyomark: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Robyn Larracy:** Methodology, Investigation, Formal analysis, Data curation, Conceptualization, Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources.

Satinder Gill: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Erik Scheme:** Writing – review & editing, Validation, Software, Project administration, Funding acquisition, Visualization, Supervision, Resources, Methodology, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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